

Semantic-World Knowledge Interactions During Sentence Processing

Background

Puzzle: The productivity & systematicity of language suggest semantics is strictly compositional...

Principle of Compositionality

The meaning of a complex expression is a context-insensitive function of the meanings of its parts^{1,2}

...But lexical context dependence effects may pose a challenge

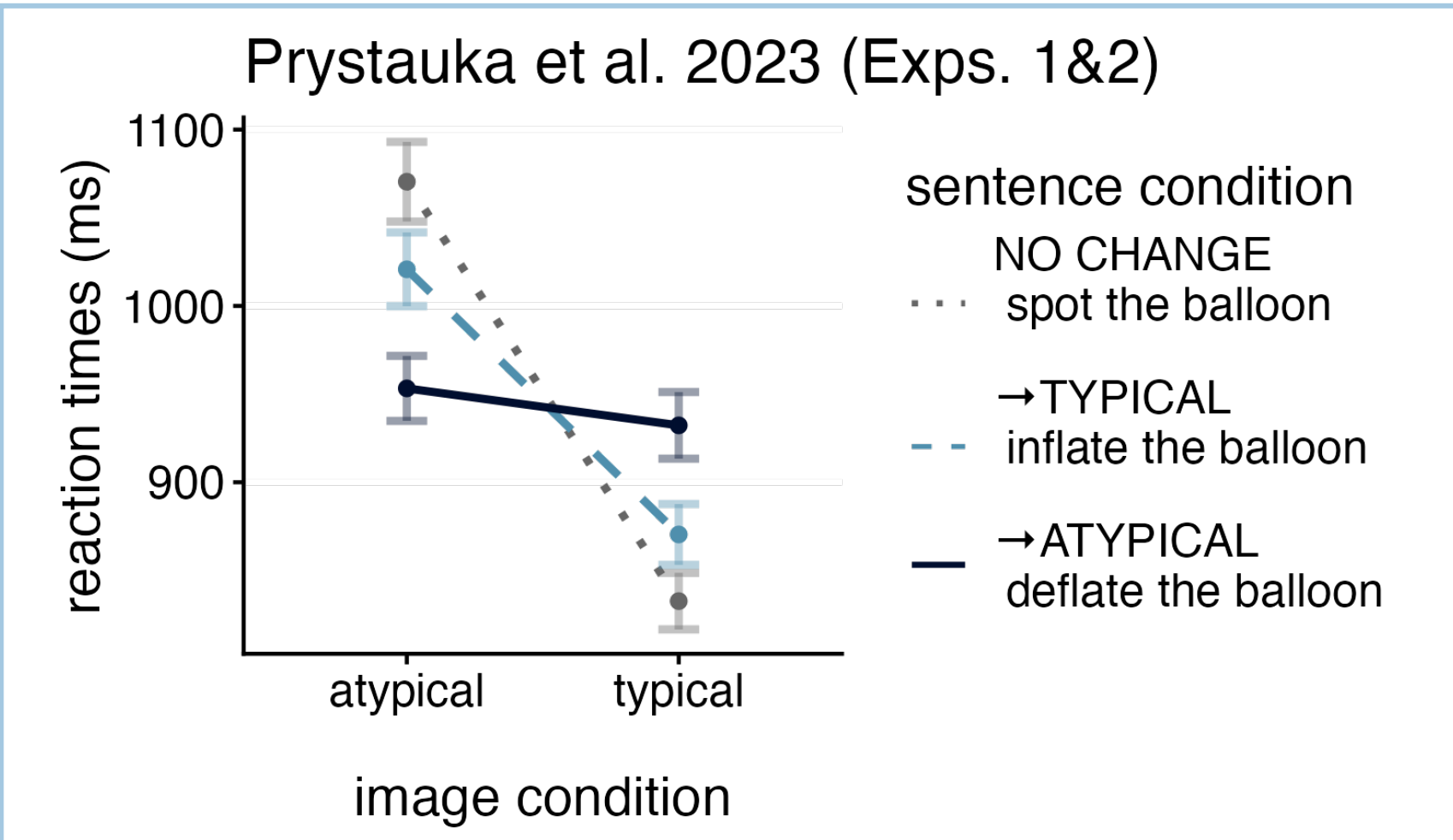
- many rules seem sensitive to world knowledge of what the units of composition refer to
- context-dependent meanings of lexical items appear to compose during processing³⁻⁸

Research Question: Many of these lexical context dependence effects are observed in offline tasks. Are world and linguistic knowledge integrated *during* processing, and if so, when?

Test case: VP → V NP

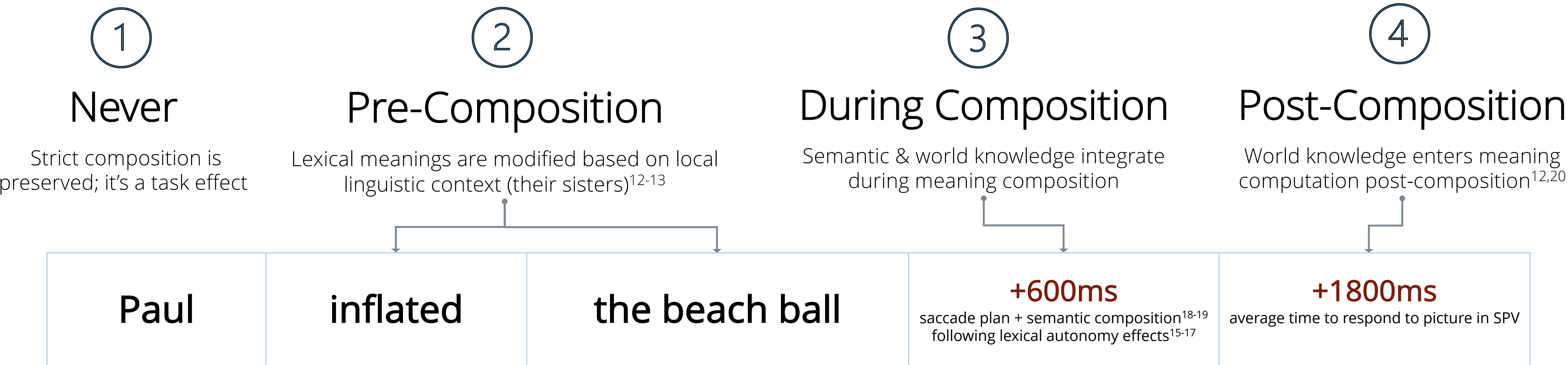
Task: Sentence-Picture Verification (SPV),⁹⁻¹¹ Eye tracking, VWP

Effect: Object typicality modulates effects of sentence.



Hypotheses & Predictions

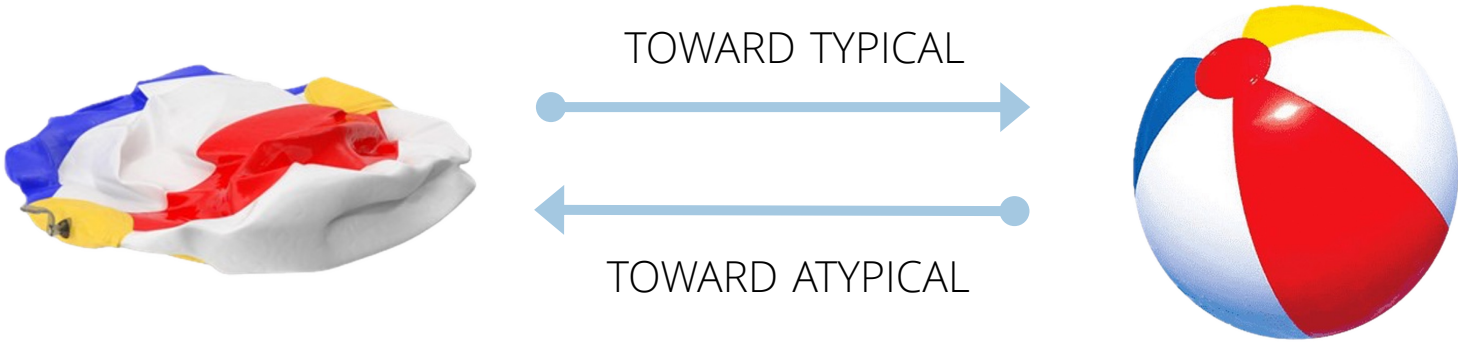
When, if ever, does an OBJECT TYPICALITY x SENTENCE CONDITION pattern arise during processing?



Methods

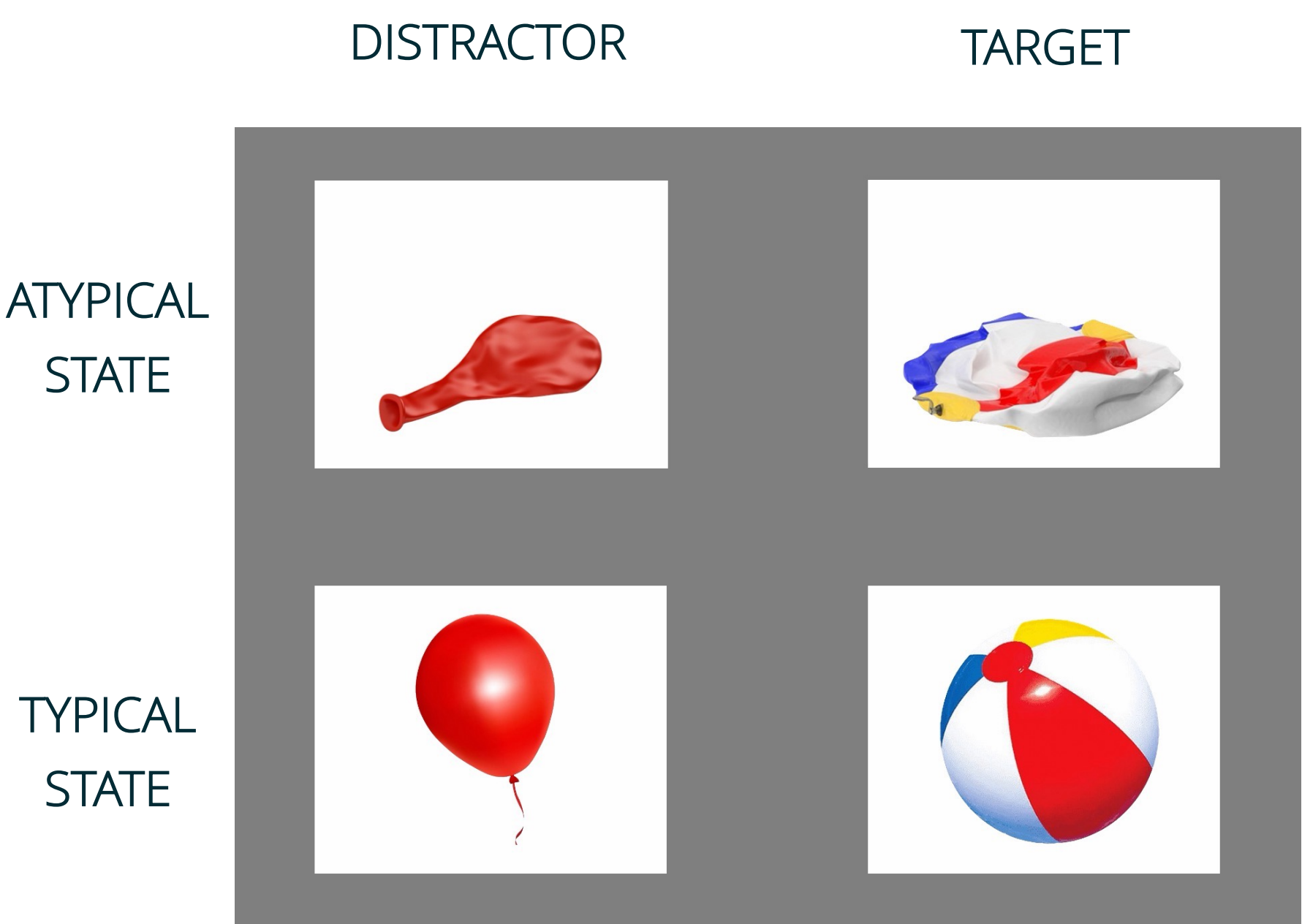
Sentence Conditions.

- (a) Paul spotted the beach ball. NO CHANGE
- (b) Paul inflated the beach ball. CHANGE TOWARD TYPICAL STATE
- (c) Paul deflated the beach ball. CHANGE TOWARD ATYPICAL STATE



Picture Conditions.

TYPE (target, distractor) x TYPICALITY (typical, atypical)

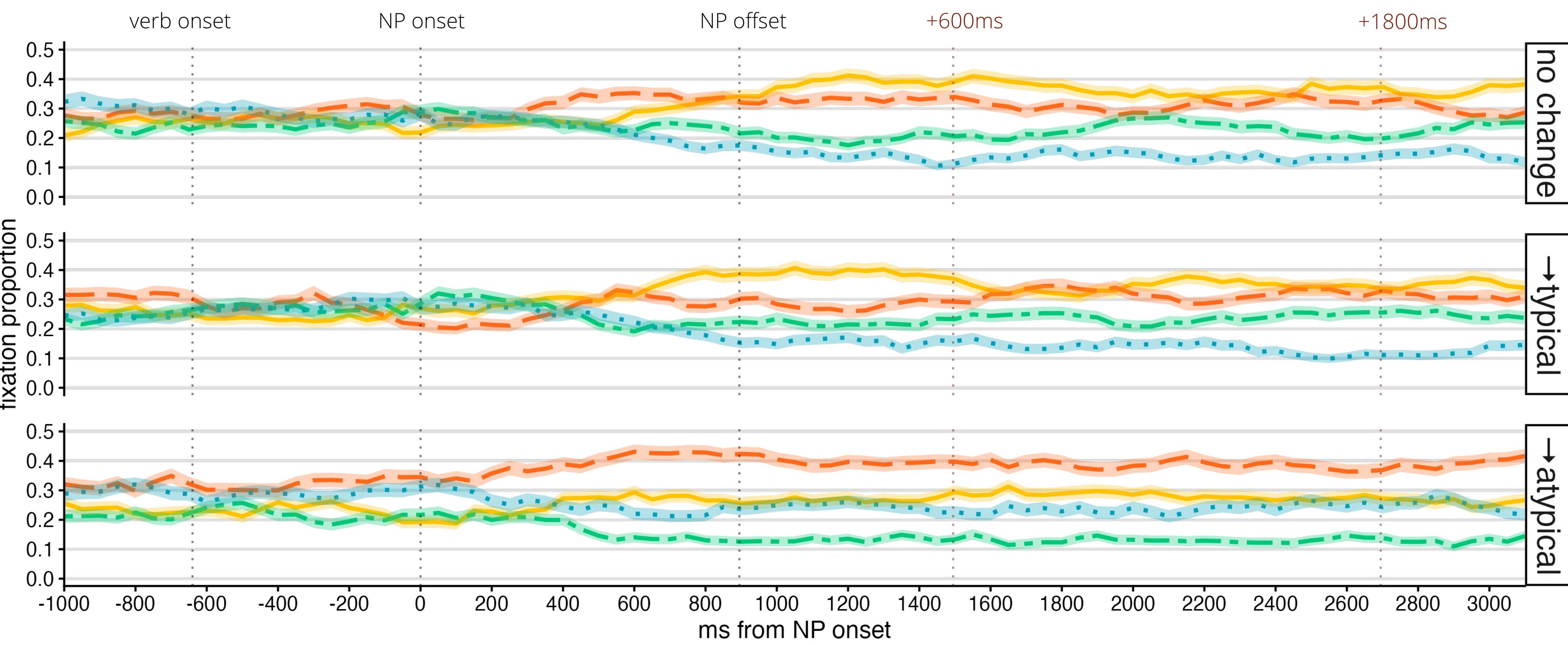
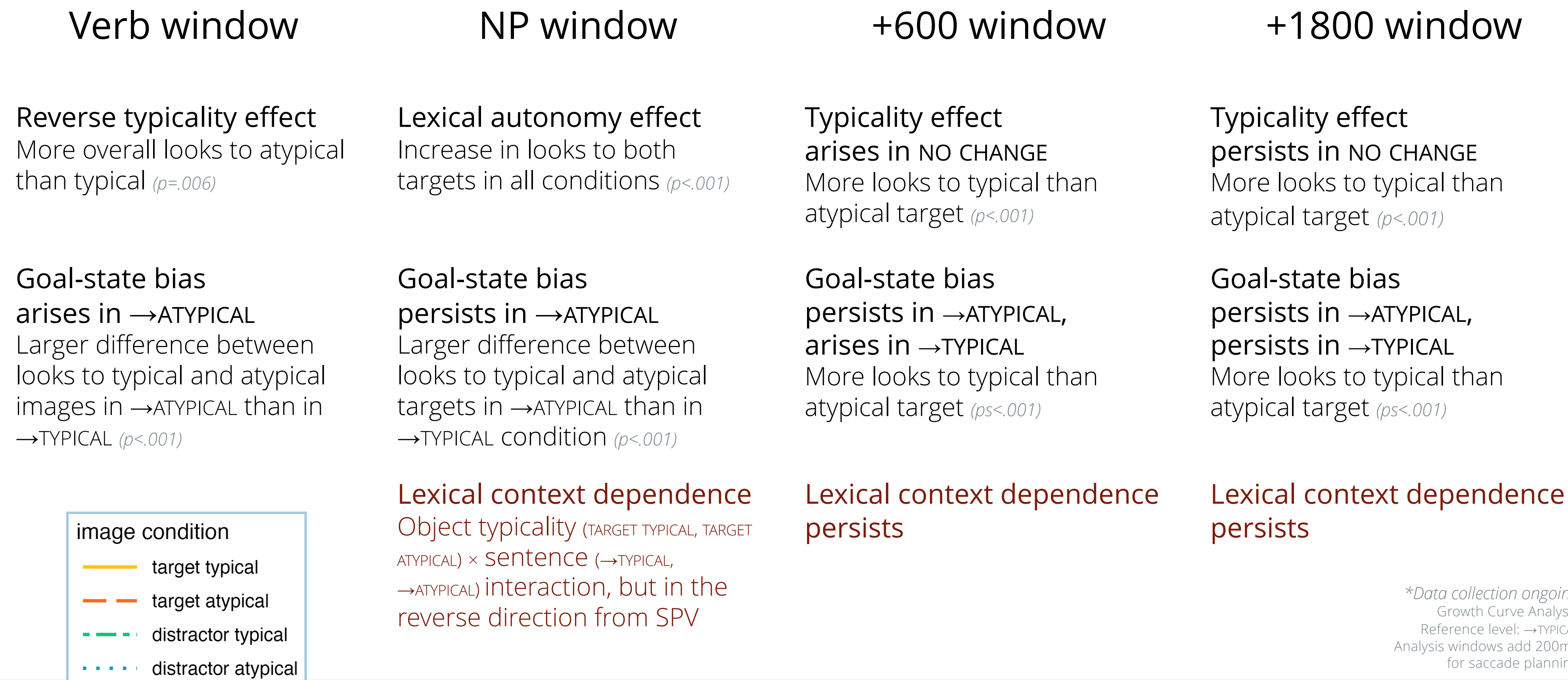


Items. 96 objects (typical & atypical state). (b)-(c) are reversible telic incremental theme verb phrases entailing external object change

Procedure. Eye tracking in the Visual World Paradigm (VWP; Eyelink 1000+ with sampling rate of 500Hz) during sentence listening

Participants. Self-reported monolingual native English-speaking UConn undergraduate students (N=26); data collection ongoing

Preliminary Results



Discussion

Initial results suggest...

A **difference** between offline and online: Typicality effects

- VWP shows no strong typicality effect in →ATYPICAL condition, in contrast with SPV⁹⁻¹¹
 - VWP instead indicates a potential visual salience effect (atypical > typical images) at verb
 - SPV may bring typicality strongly into processing during image recognition

A **similarity** between offline and online: SENTENCE x OBJECT TYPICALITY interaction

- This is a **lexical context dependence** effect, which arises in the **NP window** in the VWP
 - appears to be driven by the conjunction of a **goal-state bias** (initially arising in the verb window in the →ATYPICAL condition) plus a **lexical autonomy effect**
 - offers some preliminary support for Pre-Composition